



Fig. 13: Block diagram of a tank's FCS

showing images even during night time due to their low-light image capturing capability through thermal imaging. The sights are technically called forward looking infra red (FLIR) imagers. These are generally capable of detecting IR wavelengths in the 8 to 14 micron band and have their lineage from the targeting sights of combat aircraft. These imagers actually see (sense) the heat signature of a target rather than its visual image; even a target well-camouflaged inside foliage cannot escape from it. Because of these sights, contemporary tanks are as active in the nights as they are during day time. But tanks of the past were night-blind.

Crosshairs are generated electronically and are not engraved crosshairs; the ballistic computer keeps track of the alignment of the crosshairs. The gun gets heated heavily due to high pressure build up inside the barrel during firing process. The gun barrels tend to slightly bend due to this heating. This miniscule bending can lead to pronounced deviation at longer range. Often the gunner, using a system called muzzle reference system, can adjust the crosshair's position in his sight based on the barrel's degree of bending.

The commander's and gunner's sights are almost equally capable and independent. The tank commander's cumbersome opto-mechanical sights of the past have been replaced by a stabilised panoramic sight for day and night observation. Commanded

through a joystick, these panoramic sights can rotate 360°. The gunner's sight is also night-imaging capable but not panoramic. The gunner's sight can be enslaved to the commander's sight by the commander to see what the gunner sees. These sights find the range, speed and the bearing of a target!

With an embedded laser range finder (LRF) present in the sights, the ballistic computer gets the range of the target. The LRF flashes a narrow pulse-coded IR laser beam for a microsecond on the target area covered by the crosshairs. With laser sensors associated with the LRF, the reflected laser beam from the target is sensed. The time duration between the flashing and sensing gives the round-trip time between the tank and target. From this the ballistic computer calculates the range. The typical maximum range of laser rangefinders is 10 kilometres with an error of 10 to 20 metres at the extremities of the range.

Similarly, for a moving target, the gunner has to keep the crosshairs on the target for a second or two. During this time period, the computer flashes the LRF on the target repeatedly and extrapolates the target's course. Now the ballistic computer has the data regarding the target's range, speed and angle.

Sensors. Using wind sensors situated on the top of the turrets, the ballistic computer measures the crosswind speed. Using pendulum static anti-

lever sensors located at the centre of the turret roof, the ballistic computer calculates how much the tank is tilted due to uneven ground.

Similarly, the ballistic computer takes the speed of the tank and its course from a system called inertial navigation system (INS). (This INS is a navigation system that helps ships, aircraft and missiles to find their position in the open water and airspace, respectively. But a tank getting them is one of the strangest aspects of battle dynamics.) From the gunner's console, the gunner has to manually input the data regarding the ammunition type and some miscellaneous data.

Additional functions of FCS

The FCS has some additional functions. For instance, FCS lays and stabilises the gun and produces symbology for the gunner/commander.

Laying the gun. From the data derived from the sensors and sights, the firing solution is derived in no time. Accordingly, the ballistic computer actuates amplidyne that move the turret in the required position. Similarly, the gun is also actuated through the amplidyne. A constant feedback is maintained to ensure that the turret and gun are moved according to the desired angles.

If the tank is moving fast and the gun is kept at the required angle rigidly, even a slight bump on the road can take the gun off the correct angle. If at that particular instant a shot is fired, even a slight bump will reduce the accuracy. So the ballistic computer holds the gun loosely on the target through stabilisation. To sense the tank's movements, accelerometers are used as sensors that give an electrical output whenever the tank jerks. Immediately the FCS correspondingly controls the amplidyne and maintains the gun in the required orientation—all in just a fraction of second.

Symbology. Apart from these sensing and controlling operations, the ballistic computer generates symbologies on the display regarding various